Worksheet 16 · Percentages IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Find a percentage of an amount • Calculate percentage increase, decrease and change • Use reverse percentages to find an original amount		
Grade: / 80	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Multiplier → Increase = $\frac{100+x}{100}$ Decrease = $\frac{100-x}{100}$

% Change → $\frac{\text{Change}}{\text{original}} \times 100$

Reverse % → Original = New ÷ Multiplier

Questions *Percentages*

Question 1 (6 marks)

The table gives information about the price of gold.

	1st February 2016	1st March 2016
Price of one ounce of gold (dollars)	1126.50	1236.50

(a) Work out the percentage increase in the price of gold between 1st February 2016 and 1st March 2016
Give your answer correct to 3 significant figures.

..... %
(3)

The price of one ounce of gold on 1st February 2016 was 1126.50 dollars.
The price of gold increased by 19% from 1st February 2016 to 1st July 2016

(b) Work out the price of one ounce of gold on 1st July 2016
Give your answer correct to the nearest dollar.

..... dollars
(3)

Question 2 (4 marks)

Mary saves for a holiday each year.

In 2020 she saved a total of $720
In 2021, each month she saved $78

The total amount Mary saved in 2021 was P% more than the total she saved in 2020

(a) Work out the value of P

.....
(4)

Question 3 (8 marks)

Josh buys and sells books for a living.

He buys 120 books for £4 each.

He sells $\frac{1}{2}$ of the books for £5 each.

He sells 40% of the books for £7 each.
He sells the rest of the books for £8 each.

(a) Calculate Josh's percentage profit.

..... %
(5)

One book that Josh owns had a value of £15 on the 1st May 2019
The value of this book had increased by 20% in the last year.

(b) Find the value of the book on the 1st May 2018

£.....
(3)

Question 4 (6 marks)

Lorenzo increases all the prices on his restaurant menu by 8%

Before the increase, the price of a dessert was $4.25

(a) Work out the price of the dessert after the increase.

$.....
(3)

After the increase, the price of lasagne is $9.45

(b) Work out the price of lasagne before the increase.

$.....
(3)

Question 5 (7 marks)

A company makes cars.

In 2016, the company made 350 cars.

In the first 6 months of 2017, the company made 25 cars each month.
In the last 6 months of 2017, the company made 45 cars each month.

(a) Work out the percentage increase in the number of cars the company made from 2016 to 2017

..... %
(4)

The company's income in 2017 was $500 000 more than the company's income in 2016

The company's income in 2017 was 8% more than the company's income in 2016

(b) Work out the company's income in 2016

$.....
(3)

Question 6 (6 marks)

Gladys buys a table for $465 to sell in her shop.

She sells the table for $520

(a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

..... %
(3)

Gladys has a sale in her shop.

She decreases all the normal prices by 12%
The normal price of an armchair was $550

(b) Work out the sale price of the armchair.

$.....
(3)

Question 7 (6 marks)

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	Service charge band
$x \geqslant 700\,000$	A
$600\,000 \leqslant x < 700\,000$	B
$500\,000 \leqslant x < 600\,000$	C
$400\,000 \leqslant x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

(a) Has the annual service charge band changed for Asha's apartment?
Show your working clearly.

(3)

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

(b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %
(3)

Question 8 (5 marks)

Sarah makes and sells mugs.

One day she makes 150 mugs.
Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.
Give your answer correct to the nearest whole number.

..... %

Question 9 (3 marks)

In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

..... million

Question 10 (3 marks)

On Wednesday, the price of 1 litre of petrol was £1.26
The price of petrol on Wednesday was 5% more than the price of petrol on the previous Monday.

Calculate the price of 30 litres of petrol on the previous Monday.

£.....

Question 11 (3 marks)

Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

(a) Work out how much Chengbo paid for the house.
Give your answer correct to 3 significant figures.

..... yuan
(3)

Question 12 (5 marks)

The table gives information about the average house price in England in 2018 and in 2019

Year	2017	2018	2019
Average house price (£)		228 314	231 776

(a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

.....
(2)

The average house price in 2019 was 7.7% greater than the average house price in 2017

(b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£.....
(3)

Question 13 (4 marks)

Simon bought a house at the beginning of 2018
The value of Simon's house had decreased by 15% by the end of 2018

The house increased in value during both 2019 and 2020
The percentage increases in the value of the house during 2019 and 2020 were the same.

The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018

Calculate the percentage increase in the value of the house during 2019

.....%

Question 14 (7 marks)

Shane bought a car.

The amount Shane paid for the car was $32 000

Theresa also bought a car.
To pay for this car, Theresa paid a deposit of $18 000 together with 14 monthly payments of $1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

.....%
(4)

Kylie bought a van.

After 1 year, the value of the van was $39 865
During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

$.....
(3)

Question 15 (7 marks)

Gladys runs a furniture shop.
She buys a table for $465 and sells it for $520.

(a) Calculate the percentage profit she makes from selling this table. Give your answer correct to 3 significant figures.

.....%
(2)

Gladys has a sale in her shop and reduces all prices by 12%. The sale price of an armchair was $550.

(b) Work out the price of the armchair before the sale.

$.....
(2)

Gladys purchases a second-hand sofa for $900. There is an additional shipping fee of $75 that she must also pay.

She plans to sell the sofa so that she makes at least 15% profit.

(c) Find the minimum selling price she should charge for the sofa to ensure a profit of 15%.

$.....
(3)